

Department of Physics and Astronomy

First Year

2024-25

Tutorial Sheet 10

PHYS10302 Vibrations and Waves

PHYS10342 Electricity and Magnetism

PHYS10372 Mathematics 2

This sheet covers tutorials in weeks 11 of semester 2.

Problem solving can be a difficult skill to master. If you are finding it tough going, it may be helpful to try the following strategy:

1. Identify what the problem is about. Try to visualise the problem. Drawing a diagram may help.
2. Identify clearly what you are being asked to find.
3. What information have you been given? Sometimes even the simple task of writing down this information can help.
4. What (physics) principles and equations can you use?
5. How do you put points 1, 2, 3 and 4 together to get a solution? What maths will you need?
6. It is usually best to try and solve the problem algebraically, using symbols, even if you are given numerical values. This enables you to check at each stage and at the end, that you have correct and consistent dimensions. The last thing to do is to put numbers in, if a numerical answer is required.
7. Is the answer reasonable? If the answer is numerical, is the order-of-magnitude appropriate? Does your algebraic answer behave in the correct way when the variables take on very large or very small values, or in other limits?

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PHYS10302 Vibrations and Waves

Week 5 Questions

1. We can express the positions of two coupled masses (assuming zero initial phase angles) as

$$\begin{aligned}x_a &= \frac{1}{2} [C_1 \cos(\omega_1 t) + C_2 \cos(\omega_2 t)] \\x_b &= \frac{1}{2} [C_1 \cos(\omega_1 t) - C_2 \cos(\omega_2 t)]\end{aligned}$$

Assume $C_1 = C_2 = A$ and use the standard trigonometric identities for the cosine of a sum of two angles,

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta,$$

to express the equation for x_a as the product of two cosine functions. If the two normal mode frequencies are quite close to each other, interpret the result in terms of an oscillation at the uncoupled frequency together with a slow modulation.

2. For a pendulum, explain why we can use the distance from the equilibrium position x , instead of the angle the pendulum makes with the vertical θ , to describe the motion of the pendulum mass.

What is the period of a simple pendulum of length 0.4 m?

Two identical pendula, each of length 0.4 m, are connected by a light coupling spring of spring constant k . If one pendulum is clamped, the period of the other is 1.25 s, which includes the effect of gravity and of the spring. Show that this means that $\frac{k}{m} = 0.741 \text{ rad}^2 \text{ s}^{-2}$. With both pendulums unclamped, what are the periods of the two normal modes?

3. Consider two equal masses, suspended vertically, with the lower mass (b) and spring attached to the upper mass (a) and spring. This forms a coupled system. The spring constants are both equal to k . Displacements for both masses are measured in the downward direction.

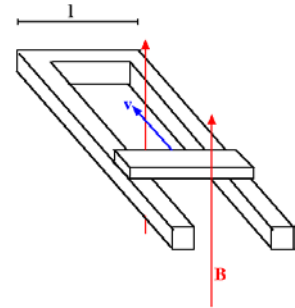
The equations of motion for the two masses are

$$\begin{aligned}m \frac{d^2 x_a}{dt^2} &= -kx_a + k(x_b - x_a) = k(x_b - 2x_a), \\m \frac{d^2 x_b}{dt^2} &= -k(x_b - x_a).\end{aligned}$$

Using $x_a = A \exp(i\omega t)$ and $x_b = B \exp(i\omega t)$ write the resulting equations as a matrix equation. Hence solve the eigenvalue problem to find the frequencies of the normal modes.

E&M Tutorial Sheet 10: Power and Energy

- 1) (Revision) A circuit consists of two conducting rails, spaced a distance l apart, which are joined by a wire at one end and a horizontal metal bar that is free to move along the rails as shown in the diagram. There is a magnetic field with strength B perpendicular to the plane of the circuit. Calculate the EMF induced if the bar moves at a speed v towards the connected end of the circuit.



- 2) Consider a voltage source that provides an EMF (or potential difference), ϕ , but instead of being an ideal power supply, it has an internal resistance, r . It is used to drive current through a load that has a resistance, R .
- What current flows through the circuit in terms of ϕ , r and R ?
Hint: the resistors are in series.
 - What is the power dissipated in the load resistor in terms of ϕ , r and R ?
Hint: power = $I^2 R$.
 - What value of R maximises this power?
Hint: Differentiate the expression from (b) and thus find the maximum.

This result is called impedance matching. Often you can't just alter the resistance of your load circuit to satisfy this condition. But if the voltage source is A.C. (alternating current) you can use a transformer.

- If a transformer is placed between the voltage source and load with N_1 and N_2 turns, what is the relationship between ϕ_1 and ϕ_2 , the potential differences across the two coils respectively?
 - What is the corresponding relationship between I_1 and I_2 ?
 - What is the effective load seen by the voltage source i.e. ϕ_1/I_1 ?
 - What can be altered to make the answer to (f) satisfy the impedance matching condition? If r was 20Ω and R was 320Ω , what turns ratio would ensure maximum power delivery?
 - Why can this method not be used for D.C. (direct current)?
- 3) The magnetic field in interstellar space is approximately 10^{-10} T. These regions of space contain around 1 hydrogen atom per cubic centimetre, which have an average speed of around 10^5 cm s⁻¹. Compare the magnetic energy per unit volume to the kinetic energy per unit volume.

PHYS10372 Mathematics 2: Tutorial-10 Questions

1. Use the Gauss-Elimination method on the augmented matrix to produce a RREF matrix that can solve for x_1, x_2, x_3 in the following equations:

$$\begin{aligned}2x_1 + 3x_2 + x_3 &= 5 \\4x_1 + 5x_2 + 2x_3 &= 8 \\3x_1 + x_2 + x_3 &= 4\end{aligned}$$

Write this equation in matrix form as: $A\mathbf{x} = \mathbf{b}$ and verify your previous solution by solving this matrix equation for $\mathbf{x}$ by inverting A .

2. Find the rank of the following matrices by forming the respective RREF matrix.

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 4 & 5 & 2 \\ 3 & 1 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 2 & 2 & 7 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 6 & 8 \end{pmatrix}.$$

Solve the equation: $D\mathbf{x} = \mathbf{b}$, for $\mathbf{x}$ where $\mathbf{b} = (5, 10, 15)^T$. What is the solution if instead $\mathbf{b} = (5, 10, 20)^T$.

3. Find the eigenvalues and eigenvectors of the matrix $\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$.

If the eigenvectors represent the displacements of three connected masses, what relative motions of the masses do the three eigenvectors describe?

4. Data is fitted to a model that has two parameters: a and b and the best-fit (minimum χ^2) returns values of $a = 2.0$ and $b = 1.0$ respectively.

The fit to the data returns the following covariance matrix: $C = \begin{pmatrix} 4 & \sqrt{3} \\ \sqrt{3} & 2 \end{pmatrix}$.

- What do each of the terms in this matrix represent? What is the correlation coefficient between the two parameters?
- On a plot of a vs b draw the curve defined by the (a, b) values where the χ^2 of the fit is one more than the best-fit χ^2 .
- By finding the eigenvalues and eigenvectors of C show that C can be expressed in the form: $C = P^{-1}DP$ where D is a diagonal matrix.
- The Cholesky decomposition matrix, L , is a lower triangular matrix of the form $L = \begin{pmatrix} p & 0 \\ q & r \end{pmatrix}$ such that $LL^T = C$.

Determine L . This matrix can be used to quickly generate pairs of (a, b) values that respect the uncertainties in a and b and the correlations between a and b .

5. How many arithmetic operations (multiplication and subtraction/addition separately) are required to determine the determinant of a 2×2 matrix using the simple cofactors method? What does this become for 3×3 and 4×4 matrices? Generalise this for the determinant of a $n \times n$ matrix.

A 4 GHz, multi-core, computer can perform 10^{11} arithmetic operations per second and requires 100W of power at an electricity cost of 25p per kWh and a daily standing charge of 50p. Estimate the time taken and the cost incurred to find the determinant of a 50×50 matrix using such a computer implementing the simple cofactor method.